



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering Academic Year 2022 - 2023 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. ECE A

Course Code & Title: EC8691 Microprocessors and Microcontrollers

Name of the Faculty member: Mr.B.Kannan, AP (SG)

Innovative Practice Description

- **Unit/Topic:** Unit-IV / Addressing Modes
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 10
- **Activity Chosen:** Hobnob Group
- **Justification:**

The addressing modes help us specify the way in which an operand's effective address is represented in any given instruction. Some addressing modes allow referring to a large range of areas efficiently, like some linear array of addresses along with a list of addresses. The 8051 addressing modes include different categories with a few instructions. Knowing the instruction belongs to the possible addressing mode is important to understand the working of the instruction and to write the code to minimize the memory requirement and to utilize the controller properly without delay. Hence, students will understand the system design with using microprocessor with peripheral devices. This hobnob group activity is intended to help students remember what they've learned and improve their learning through teamwork.

- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**

(i) Materials for the Activity:

The topic for the groups was distributed at the time of the activity's proceeding. I informed them of the activity date two days before and asked the students to bring a hard copy of the material for their preparation. Students were also required to refer to the web content in addition to the material.

(ii) Formation of Groups:

As per the number of sub-topics, student groups were created as below:

- Number of Topics: 06
- Class Strength: 48
- Number of groups: 06
- Members per group: 08

(iii) Topic Allocation:

For six sub-topics, students were divided into six groups, and topics were allotted on the spot. The student's group discussed the topic in detail for 20 minutes with the materials.

(iv) **Topic Delivery:**

From each group, one person joined the other group to disseminate the topics learned. The students have to answer questions from other group members.

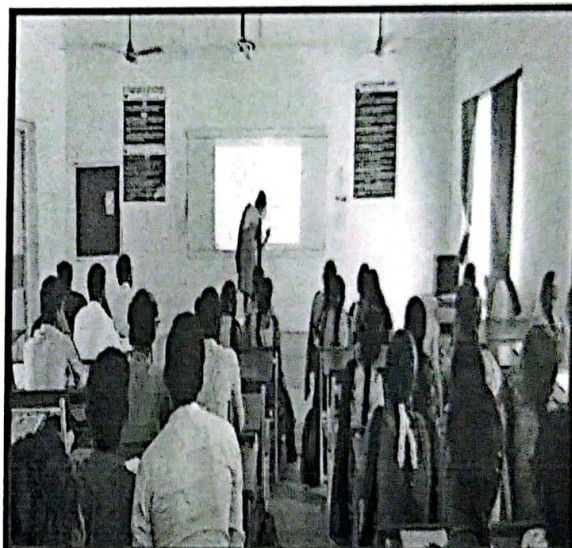
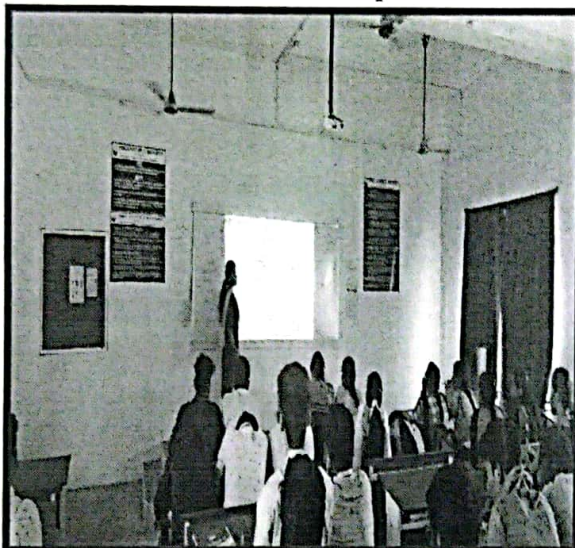
• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO9	PO10	PO11	PO12
C303.4	3	2	3	1	1	-	-	1	1	1

• **PO / PSO mapped:**

Innovative practice	PO9	PSO2
	1	2
Justification for correlation	As team students prepare a topic for the innovative activity, it enables the students to learn and diverse the content as an individual and as a team. Thus, the course outcome is mapped at level 1.	The real-time system helps to design a microprocessor system to satisfy the end-user. Hence, the course outcome is mapped at level 2.

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback on practice from students and other stakeholders:**

- This activity helps to share the topics simultaneously among all groups.
- They believed that this form of exercise boosted their conscientious abilities.
- It aids in the development of effective communication skills and active learning.

❖ **Benefits of the practice:**

- It created an opportunity for the students to deliver their topics to other members of the group.
- Knowledge sharing on different topics within 50 minutes is possible through this activity.
- All students actively participated without hesitation.

❖ **Challenges faced in implementation:**

- A few students hesitated to express the learned topics because of the group shifting from parent groups to other groups.
- Extra time was taken to rearrange themselves to form a new group.
- A few groups asked for extra time for preparation.

• **Organising a Hobnob activity:**

Step 1: Divide the class into 'expert' groups of four to six learners (keeping the groups the same size as far as possible). Ask the learners within each group to number themselves one to four (or one to six).

Step 2: Give each group a section of text or information. The group should spend some time reading, discussing and helping each other to understand the text or information.

Step 3: Learners, who are now 'experts' on the own section of text or information, then move into 'Hobnob' groups, with a shared number, i.e. all the number ones work in a group, all the number twos work in another group, etc.

Step 4: Each 'expert' learner in turn shares with their 'Hobnob' group the section of text or information they were originally given. The others ask questions to ensure all have a good understanding.

Step 5: The 'Hobnob' group together complete a task which requires them to understand all of the information shared by each 'expert'. This could be anything that requires each learner to contribute their piece of expert knowledge: filling in a grid or table, completing a diagram, designing a poster, devising a role play.

• **References:**

Book Title/ Author/ Publisher/ Edition
❖ Yu-Cheng Liu, Glenn A.Gibson, "Microcomputer Systems: The 8086/8088 Family - Architecture, Programming and Design", Second Edition, Prentice Hall of India, 2007.
❖ Mohamed Ali Mazidi, Janice Gillispie Mazidi, Rolin McKinlay, "The 8051 Microcontroller and Embedded Systems: Using Assembly and C", Second Edition, Pearson Education, 2011.
Websites
❖ https://nptel.ac.in/courses/117104072
❖ https://www.electronicshub.org/8051-microcontroller-assembly-language-programming/


Signature of Faculty Member

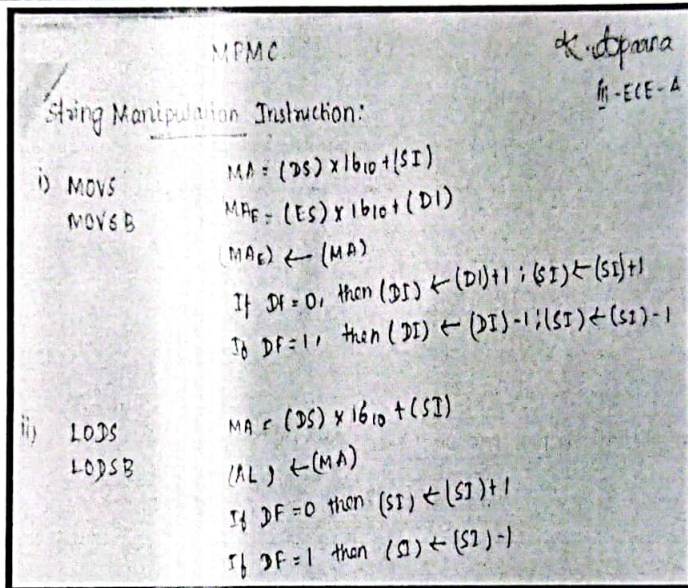
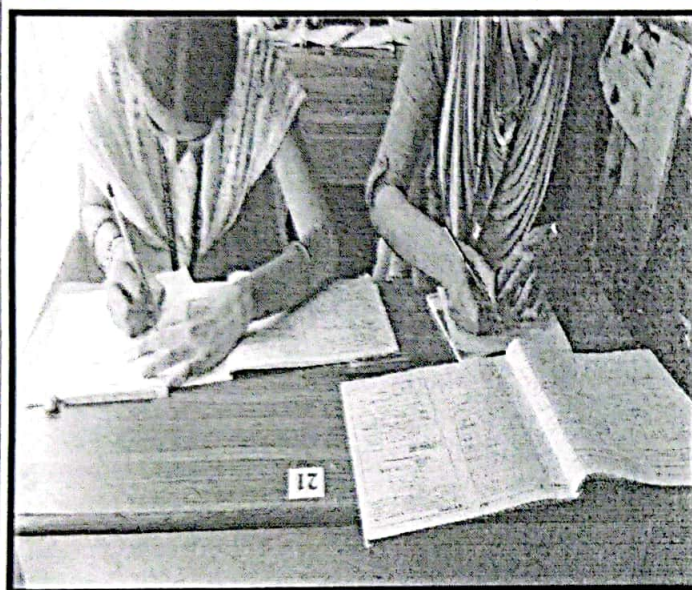

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RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 20202-2023 (Even Semester)
Active Learning

Degree, Semester & Branch: VI Semester B.E. ECE A
Course Code & Title: EC8691 Microprocessors and Microcontrollers
Name of the Faculty member: Mr.B.Kannan

Name of the Topic	Name of the Innovative Practice	Date & Duration
Byte and String manipulation Instructions	One Minute Paper	27.02.2023 (10 Minutes)

- A "one-minute paper" may be defined as a very short, in-class writing activity (taking one-minute or less to complete) in response to an instructor-posed question.
- It prompts students to reflect on the day's lesson and provides the instructor with useful feedback.

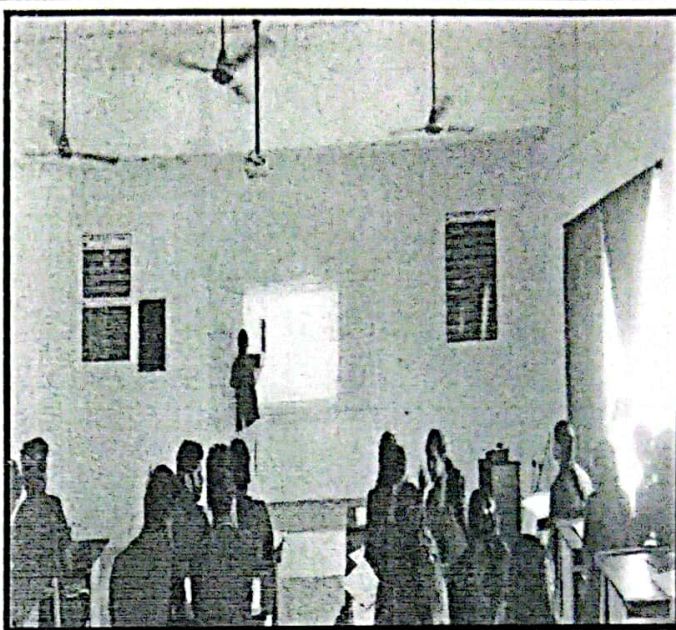
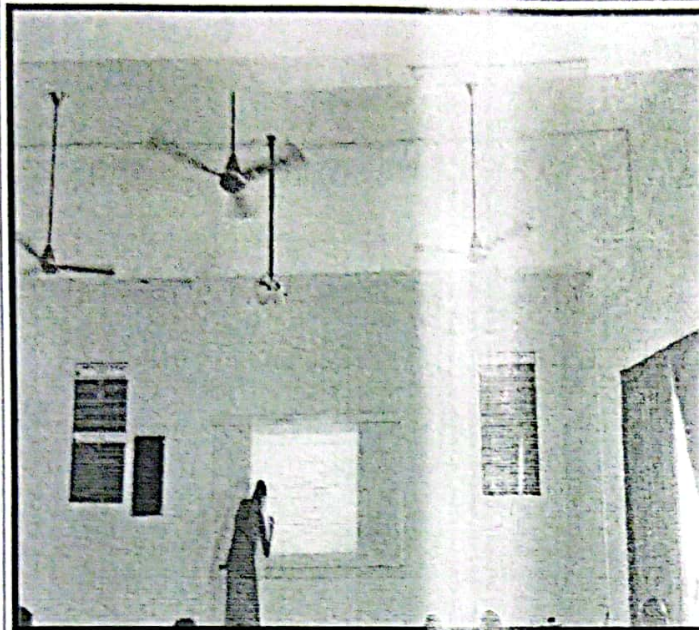


The 8086 Minimum Mode Configuration	Fast Quiz	01.03.2023 (10 Minutes)
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- A fast quiz refers to a short test of knowledge; typically around 10 questions from the block diagram.
- Helps to remember the different blocks available in the block diagram and the bus connection to interact with other peripherals.



- Mindmapping is a strategy that helps students study and teaches course material. A mindmap is a diagram that is used to visually outline information.



References:

- ❖ <http://www.icaltefl.com/just-a-minute-speaking-activity>
- ❖ <https://busyteacher.org/20564-5-steps-to-making-fluency-fun-just-a-minute.html>
- ❖ <https://oncourseworkshop.com/self-awareness/one-minute-paper/>
- ❖ <https://www.edutopia.org/blog/design-thinking-one-minute-papers-ashley-nahornick>
- ❖ Yu-Cheng Liu, Glenn A. Gibson, "Microcomputer Systems: The 8086 / 8088 Family - Architecture, Programming and Design", Second Edition, Prentice Hall of India, 2007.

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